Kinematics, dynamics and control of robots - Homework project 1

All calculations and programing should be done **parametrically**. Numerical values should only be substituted for creating the figures.
Figure 1 shows a spatial serial manipulator with 6 DOFs - $\theta_1, \theta_2, d_3, \theta_4, \theta_5, \theta_6$

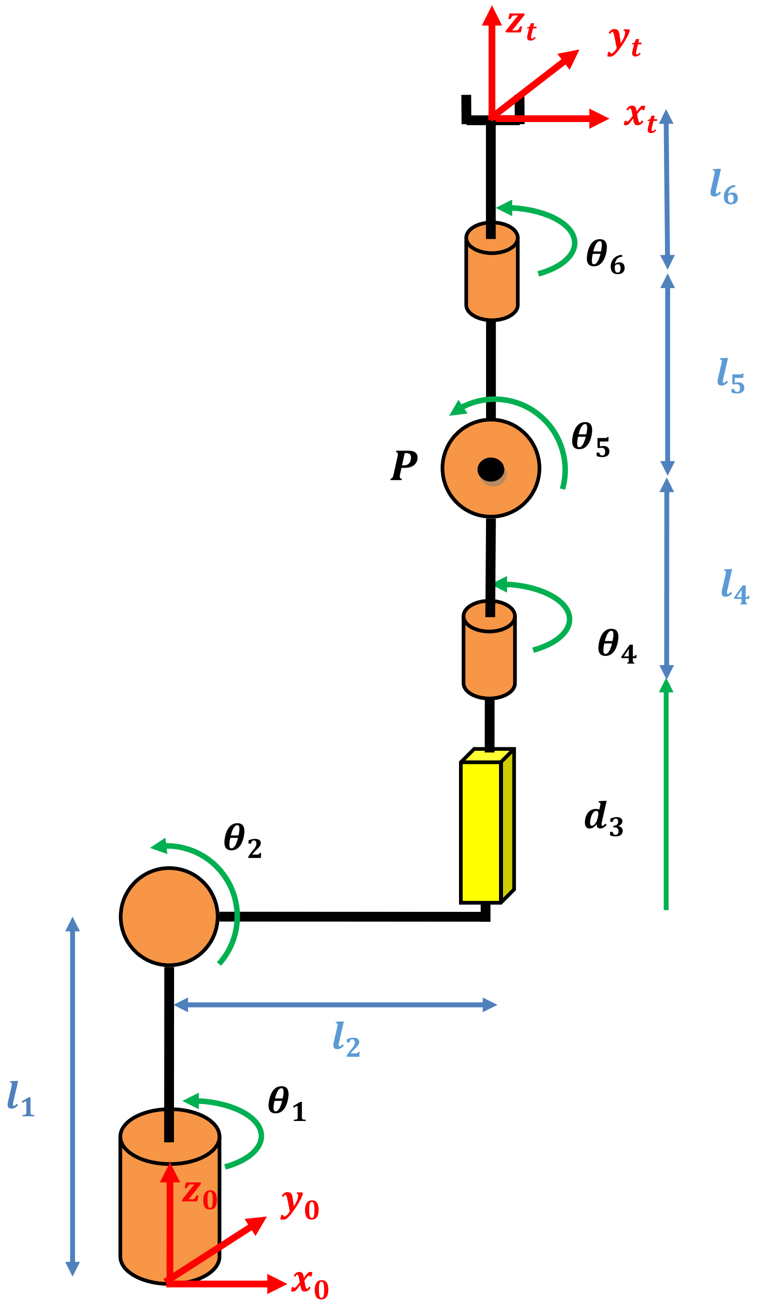


Figure 1: The spatial serial manipulator in his zero position

1. Solve the forward kinematics problem for the robot – calculate the homogeneous transformation matrix (4×4) from the tool frame $(\hat{\boldsymbol{x}}_t, \hat{\boldsymbol{y}}_t, \hat{\boldsymbol{z}}_t)$ to the world frame $(\hat{\boldsymbol{x}}_0, \hat{\boldsymbol{y}}_0, \hat{\boldsymbol{z}}_0)$ as a function of the joint variables.
   - The world and tool frames are defined in the figure 1.
   - The revolute joints are drawn in their zero position.
   - The positive directions of the joint angles are as described in the drawing.

2. Solve the inverse kinematics problem – given the 4x4 homogeneous transformation matrix representing a possible position and orientation of the robot tool, calculate the values of the joint variables.
   Find all possible solutions and show the multiple solutions in a qualitative drawing.
   Hint - an intermediate point $\boldsymbol{P}$ can be used (see Figure 1), solve the first three joints by the position of $\boldsymbol{P}$, the last three joints by the orientation problem of the end effector.

3. Calculate the full Jacobian matrix (6×6) for the robot in the world frame and the tool frame.

4. Plan the motion of the tool from point and orientation A to B (described in next page) over 2 seconds, considering the mechanical limitations of the joints (see next page). The position of the end effector should move along a straight line connecting positions A to B. The orientation of the end effector should rotate between initial orientation A to final orientation B, as described in Fig. 2. Plan the trajectory using the following time profiles of the linear and angular velocities:
   - Constant velocities.
   - Trapezoidal velocity profile – constant acceleration to maximal velocity, motion at constant velocity, and constant deceleration to stop. The time of acceleration and deceleration should be equal and the motion at constant velocity will be 2/3 of the total time.
   - Polynomial velocity profile that assures zero velocity and acceleration at the start and end of the motion.

For each of the motion profiles, present the following:

a. Analytic expressions for the position, velocity and acceleration of the end effector as a function of time.
b. Analytic expressions of the orientation matrix of the end effector as a function of time.
c. Figures of position, linear velocity and acceleration of the end effector as a function of time.
d. Figures of the joints variables (in units of degrees or meters accordingly), their velocities, and accelerations as functions of time.
   The velocities and accelerations of the joints should be calculated in two methods (and presented on the same figure):
   1. Numeric differentiation of the joint positions $\boldsymbol{q}(t)$.
   2. From the relation $\boldsymbol{v} = J\dot{\boldsymbol{q}}$ (where J_L is the linear part of the Jacobian matrix) for the velocities and $\dot{\boldsymbol{v}} = \dot{J}\dot{\boldsymbol{q}} + J\ddot{\boldsymbol{q}}$ for the accelerations.

   A list of functions to be submitted is in the following page.

For the following sections assume $\boldsymbol{\theta_4, \theta_5, \theta_6 = 0}$ and $\boldsymbol{l_4 = 0, l_5 = 0, l_6 = 0}$, so that the robot contains only the first three joints

5. Find all the singular states of the robot with respect to the task of tool position, sketch the states showing the singular direction.

6. Find the forces and torques in the joints, considering pointed mass $\boldsymbol{M}$ held by the gripper, gravitation on $-\hat{\boldsymbol{z}}_0$ direction, while masses of links and joints are negligible.

   Submission in pairs by 29.04.2025 on the course moodle page.

<u>Numerical values for computer simulation:</u>

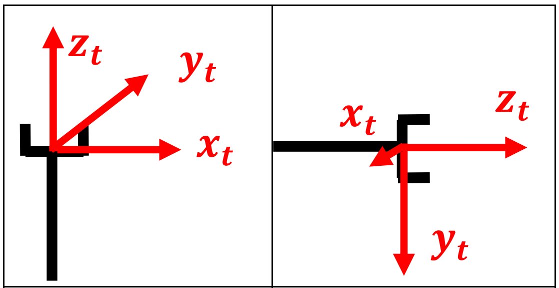
Link lengths:

$l_1 = 0.4[m]$, $l_2 = 0.15[m]$, $l_4 = 0.1[m]$, $l_5 = 0.3[m]$, $l_6 = 0.2[m]$

Initial and final positions (in meters):

$$x_A = 0.25 \quad y_A = 0, \quad z_A = 1.1$$
$$x_B = 0.55, \quad y_B = -0.4, \quad z_B = 0.6$$

Tool orientations:



A- Initial	B- Final

Mechanical limitations:

$$-\pi \le \theta_1 \le \pi, -\pi \le \theta_2 \le \pi, \quad |d_3| < 0.7[m]$$
$$-\pi \le \theta_4 \le \pi, -\pi \le \theta_5 \le \pi, \quad -\pi \le \theta_6 \le \pi$$

Functions that need to be programed and submitted (Matlab or Python):

1) [R,d]=forward_kin(q)

   q –vector of the joints' parameters

   d – position vector of the tool

   R- orientation matrix of the tool

2) q=inverse_kin(d,R,elbows)

   q –vector of the joints' parameters

   d - position vector of the tool

   R- orientation matrix of the tool

   elbows – decision values vector for the different solutions,
   1 for elbow up, -1 for elbow down.

3) J=jacobian_mat(q)

   J - the full Jacobian matrix.

4) J_dot=jacobian_mat_dot(q,q_dot)

   J_dot – the jacobian time derivative.

   q_dot – the joins' parameters time derivative.

5) x=x_plan(prof,t)

   prof – decision value for the different velocity profile.

   x – the position of the tool's origin in time t.

6) v=v_plan(prof,t)

   v – the velocity of the tool's origin in time t.

7) a=a_plan(prof,t)

   a – the acceleration of the tool's origin in time t.

8) q=q_plan(prof,t)

   q – the joints parameters in time t.

9) q_dot=q_dot_plan(prof,t)

   q_dot – the joints velocity in time t

10) q_dot2=q_dot2_plan(prof,t)

   q_dot2 – the second time derivative of the joints' parameters vector.

• It can be useful for you if the functions 5-10 could work with vector time t and give a suitable vector/matrix output.
• It can be useful if function 1 and 2 could work with suitable vector/matrix as number of vectors q and x.
•You may add functions and make some changes to the functions without changing the general concept .
• Vectors x,q,q_dot,q_dot2,v,a are row vector (or matrix if the function work with time vector).
•In the start of every function add a short description of the function use.
•Useful Matlab functions: diff/gradient, atan2 ^., /. ,*. ,.

There should be no symbolic calculations or variables in any of these functions!